

you are given an array of n-1 integers, each integer is in the range from 1 to n with no duplicates and exactly one number is missing, identify that number

2 inputs

n = 8

from 1 to 8 ==> 1, 2, 3, 4, 5, 6, 7, 8

n = 6

1 to 6 ==> 1, 2, 3, 4, 5, 6

n = 6

arr[] = [1, 2, 4, 5, 6]

n = 4

arr[] = [1, 3, 2]

4

n = 4 ==> 1, 2, 3, 4 → 10
arr = [1, 2, 4] → 7 -

ANSWER ==>

n = 6 ==> 1+2+3+4+5+6 ==> 21
arr = [1, 2, 3, 4, 6] ==> 16 - 5

- 1) By subtraction of actual sum with expected sum
- 2) By using XOR(^)

XOR ==> if both the bits are same ==> 0

XOR ==> if both the bits are different ==> 1

10 ^ 4

1010
0100

1110
14 ==> E

int n = 7;

int arr[] = {1, 4, 2, 3, 6, 7};

[1, 2, 3, 4, 5, 6, 7]

[1, 4, 2, 3, 6, 7]

{1, 4, 2, 3, 6, 7};

0 1 5 7 4 2
1 5 7 4

[1, 2, 3, 4, 5, 6, 7]

0 1 3 0 4 1 7
1 3 0 4 1 7 0

3) Using HashSet

int n = 5;

int arr[] = { 3, 4, 1, 5 };

[1, 2, 3, 4, 5]

3, 4, 1, 5

Leaders in the Array

arr = [16, 17, 4, 3, 5, 2]

[2, 5]

Collections.reverseOrder()

n = 5

arr = [1, 2, 3, 5, 3]

find the missing and duplicay ele

[10, 20, 30, 40, 50]

k = 1

[20, 30, 40, 50, 10]